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Model Answers and Mark Schemes — First Examiner: J. BARRIA

Paper Code: EE 3.17 — Second Examiner: C. Ling

DEC 2018

Question Number etc. in left margin | Mark allocation in right margin

1a) i) Stop and wait $\quad u = \dfrac{1-P}{1+2a}$

Selective Reject

$$u \begin{cases} 1-P & W \geq 2a+1 \\ \dfrac{W(1-P)}{2a+1} & W < 2a+1 \end{cases}$$

Go back N

$$u \begin{cases} \dfrac{1-P}{1+2aP} & W \geq 2a+1 \\ \dfrac{W(1-P)}{(2a+1)(1-P+WP)} & W < 2a+1 \end{cases}$$

2

ii)

	a=0.1	a=1.0	a=10.0	a=100.0
S&W	0.8325	0.333	0.047	0.004
SR(50)	0.999	0.999	0.999	0.245
GbN(50	0.998	0.997	0.959	0.233

3

Discuss
- performance of S&W
- Compare performance of SR vs GbN
- Discuss window size impact on the performance.

3

ii) Assumption: Ack not corrupted, negligible size — 1

Approx: ($N_R = (1-P+KP)/(1-P)$) for Go back N — 1

$K \sim 2a+1$ if $W > 2a+1$

$K \sim W$ if $W < 2a+1$

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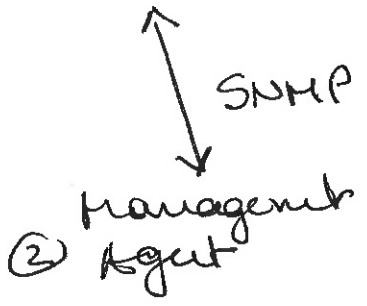
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Q1 a) i) Introduce all the following Managet elements:

Managip

① Server



5

ii) SNMP = Simple Network Managemt Protocol

- Common usage: request-response mode: the managip server send a request to an SNMP agent who will perform some actions and send a reply (MIB modif).
- Unsolicited messages are also possible: an SNMP agent can notify a managip server (link interface down).
- SNMP work on top of UDP.

3

iii) ICMP = Internet Control Message Protocol

- to communicate network-layer information between hosts and routers
- ICMP work on top of IP

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2a) i) Sources of packet delays

- Nodal processing delays
  e.g. scheduling and routing
- buffer / queueing delay
- Transmission delay
- propagation delay — 3

Nodal

- server in a queueing system
- priority queue system
- Weighted fair queue system — 3

ii) Average network delay

$N = \gamma T$

$$T = \frac{1}{\gamma} \sum \frac{F_i}{C_i - F_i}$$

γ = total/external arrival of packets into network

F_i = flow carried by link i [bps]

C_i = capacity of link i [bps] — 3

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2h) N×N switch (time slotted)

i) Since arrivals are uncorrelated the number of active time slots, K, bound for a particular output is a discrete R.V. with prob distrib:

$$P(K=k) = \binom{N}{k}\left(\frac{P}{N}\right)^k\left(1-\frac{P}{N}\right)^{N-k} \qquad k=0,1,\ldots N$$

One active time slot is lost if there are two arrivals to the same output; two active time slots are lost if there are three arrivals to the same output...
Hence the average number of lost active time slots $\langle L\rangle$ is given by 5

ii)
$$\langle L\rangle = \sum_{k=2}^{N}(k-1)\binom{N}{k}\left(\frac{P}{N}\right)^k\left(1-\frac{P}{N}\right)^{N-k}$$

$$\langle L\rangle = \sum_{k=0}^{N}(k-1)\binom{N}{k}\left(\frac{P}{N}\right)^k\left(1-\frac{P}{N}\right)^{N-k} + \left(1-\frac{P}{N}\right)^N$$

$$\langle L\rangle = N\left(\frac{P}{N}\right) - 1 + \left(1-\frac{P}{N}\right)^N$$

$$= P + \left(1-\frac{P}{N}\right)^N - 1$$

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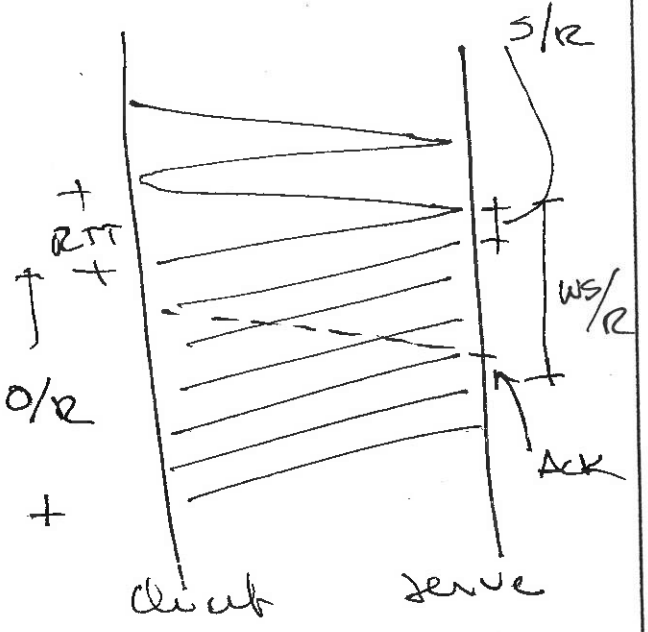
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3a i) If $WS/R > RTT + S/R$

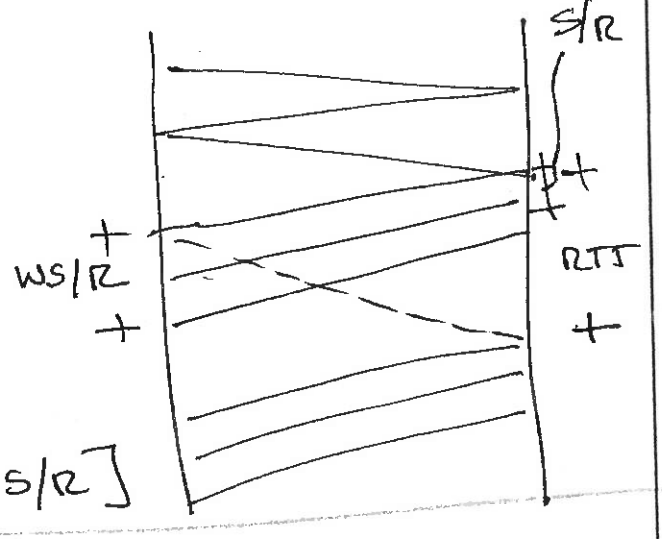
Latency = $2RTT + O/R$



2

If $WS/R < RTT + S/R$

Latency =



$2RTT + O/R$
$+ (K-1)[S/R + RTT - WS/R]$

2

ii) K = number of windows of data ($= O/WS$) 3

i) Delay = $(K-1)[S/R + RTT - WS/R]$ 2

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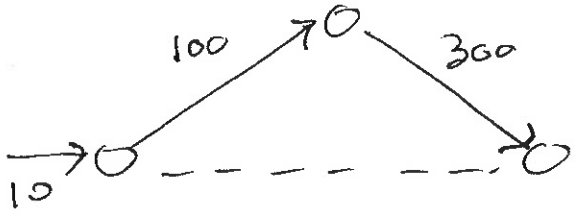
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3b

i)



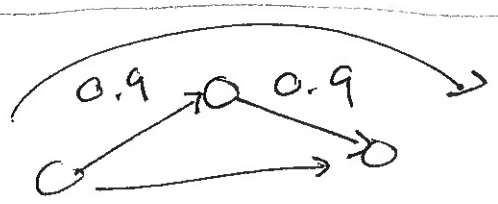
$$\frac{100}{(100-10)^2} + \frac{300}{(300-10)^2} = \frac{1}{C} \qquad 2$$

$$1.234\times10^{-2} + 3.567\times10^{-3} = \frac{1}{C}$$

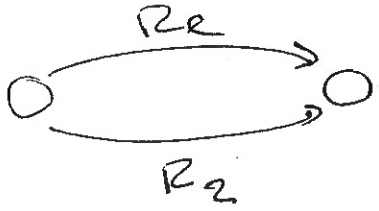
$$0.0159128 = \frac{1}{C}$$

$$C = 62.84 \qquad 2$$

ii)



$R_e = (0.9)^2 = 0.81$ 2



$$0.99 = 1-(1-R_2)(1-R_e)$$

$$= 1-(1-R_2)(1-0.81)$$

$$= 1-(1-R_2)(0.19) \qquad 2$$

$$\frac{0.99-1}{0.19} = -(1-R_2)$$

$$-0.0526 = -(1-R_2)$$

$$R_2 = 1-0.0526 = 0.947 \qquad 2$$

Assumption: Independent link failure 1

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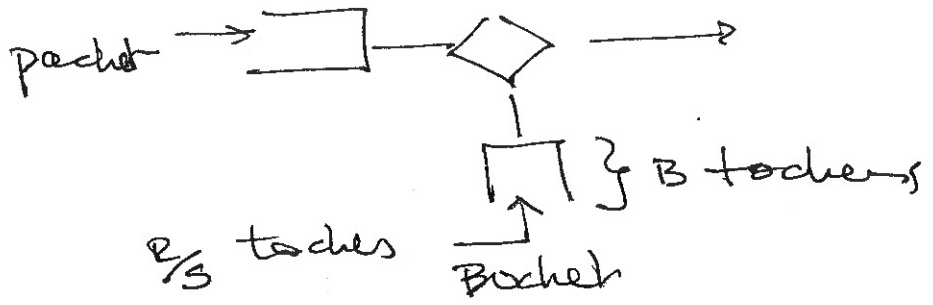
4a) Weighted Fair Queueing

i) - generalised Round Robin mechanism
- each traffic class gets weighted amount of service in each cycle.
- combine priority queueing with a buffer for each traffic class

3

Token buffer
- limit the input rate to a specific burst size and average rate

ii)



3

- bucket can hold max of B tokens
- tokens are generated at a constant rate R unless buffer is full

iii) The sender send a message containing the Traffic specification (TSpec) to receive
Receiver looks at TSpec, identify the path and initiate the procedure to reserve resource
The reservation of resource in each propagate
The RESV message contain RSpec (service request from network)

5

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4b

IEEE 802.11 (WiFi)

Application
- Connecting equipment confined to western site
- Access to WAN network

Strength
- easy deployment
- low cost

Weakness
- only within western site
- security issues

3

IEEE 802.15.4 (ZigBee)

Application
- drive-by meter reading
- user interface
- connection of sensors and other eg. in web. WAN

Strength
- low power requirement
- low cost
- good scalability
- Industrial & home automation or security appl.

Weakness
- limited range
- low data rates
- more secure than other standard

3

IEEE 802.15.1 (Bluetooth)

Applicable
- As there in 802.15.4

Strength.
- many products available
- permits higher data rates than ZigBee

Weaknesses
- limited number of devices in a network
- security issues

3